

System Documentation

Introduction

The Intervention Priority App is a quantitative decision-support system built for Reneal International's school lab network across Tanzania. Reneal operates computer labs in 120 secondary schools spanning 12 regions and over 24 districts.

With a network of this scale, field officers cannot reasonably evaluate all 120 schools through observation or memory alone. Some labs are running well, some are quietly dormant, and many fall somewhere in between — and the difference between these states is rarely visible until someone visits in person. The Intervention Priority App exists to close that visibility gap before a single visit takes place, using the usage data the labs themselves already generate.

Here specifically the Expected Hours is used to represent the average usage in number of hours of 3 near-by schools and Gap hours which represents Expected Hours - Actual Hours. Severity Multiplier means $\text{Gap (hrs)} \div \text{Actual (hrs)}$.

Summary

At its core, the Intervention Priority App takes raw lab usage hours recorded across all 120 schools from 2019 to 2025, compares each school's most recent usage against a peer benchmark drawn from similar schools in the same district, and produces a ranked priority list of the schools most urgently in need of intervention.

The system is organized into four tabs, each answering a different operational question:

Priority Rankings ranks all underperforming ("lagged") schools by a combined Priority Score, sorted from most to least urgent, with full filtering by region and field team member.

Trajectory Analysis Trajectory is determined by comparing 2023 → 2024 → 2025 usage trends. Schools with fewer than 2 data points are flagged as **Insufficient Data**. Direction and rate-of-change are calculated from available years only.

Lagged Schools Comparison Each lagged school is matched to its **3 nearest high-performing schools** (Good level, ≥ 2000 hrs actual usage) using GPS Haversine distance. The

field team officer assigned to the lagged school should arrange a peer visit to understand what those nearby schools are doing differently.

Network is the unfiltered reference view — all 120 schools, lagged and high-performing alike, with year-by-year usage history for anyone who needs the complete picture rather than just the priority subset.

Across all four views, every school is scored using the same four core metrics: Actual hours, Expected hours, Gap hours, and the $\text{Gap} \div \text{Actual}$ severity multiplier — combined into a single weighted Priority Score that determines whether a school lands in the P1 Urgent, P2 High, P3 Medium, or P4 Monitor tier.

Purpose

The Intervention Priority App was built to answer one specific operational question that Reneal's field team faces every planning cycle: **given limited time and a limited number of team members, which schools needs certain accessments, and why?**

The system serves specific purposes:

For operations, it converts raw usage logs into an ordered action list, so officers team members can see exactly which of their assigned schools carry the most lost lab time and the most severe dormancy, without needing to manually cross-reference spreadsheets.

For programme management, it provides a defensible, repeatable methodology for resource allocation — which lab need more laptops, which labs need reallocations and where.

Critically, the system was also designed with a clear boundary in mind: it tells you **where** to look and **how urgently**, but it deliberately does not try to tell you **why** a school is underperforming or **how hard** it will be to fix. That diagnostic work — assessing leadership engagement, hardware condition, teacher adoption, governance — remains the job of the team officer doing the assessment in person. The system hands them a prioritized list; it does not replace their judgment once they arrive.

Methodology

For each school, the dataset gives a multi-year usage history. Not every school has data for every year — some installations are recent (2024–2025 only), others have a long history back to 2019. The system always works with whatever years are actually available for a given school.

1. Actual (hrs) — How It's Derived

Actual is the school's most recent verified usage figure. The system applies a "most recent available year" rule:

```
Actual = 2025 hours, if recorded
        → else 2024 hours, if recorded
        → else 2023 hours, if recorded
        → else 0 (no recent data)
```

This matters because some schools have a 2025 figure, others have no records after 2024, and a few only have 2023 as their latest verified year. Rather than forcing every school onto the same year (which would either exclude schools with gaps or use stale 2019 data for schools with recent installations), the app always anchors on the *freshest available signal* for that specific school.

2. Expected (hrs) — How It's Derived

The core rule is: **Expected hours = average usage of the top 3 highest-performing (Good level, ≥2,000 hrs) schools within the same district.** Example below

Districts with 3+ Good Schools — Pure Average of Top 3

District	Top 3 Good Schools Used	Avg of Top 3	Expected Set To
Arusha CC	Mkonoo (6,784) · Arusha Day (3,224) · Themí (2,475)	4,161	3,200 ¹
Arusha DC	Einoti (21,162) · Losinoni (10,180) · Kiranyi (3,555)	11,632	1,932 ²
Butiama DC	Kiagata (10,362) · Kiabakari (6,575) · Bumangi (5,292)	7,410	5,500 ³

Expected hours is a peer-benchmark, not a target set by Reneal. It answers the question:
"Given the schools around this one geographically and administratively, what level of lab usage should we realistically expect here?"

The Method: District Near-by Level Peer Averaging

The expected value is calculated using a **K-Nearest-Neighbours (KNN) style approach**, but instead of using raw GPS distance for every single school, the app groups schools by **district** (e.g. Arusha DC, Kondoa DC, Longido DC) and computes a peer benchmark from the highest-performing schools within that same district cluster.

Expected (hours) = average usage of the top-performing reference schools
within that same near-by area

For example:

- All schools in **Kondoa DC** (Amani Abeid Karume, Bukulu, Bicha, Ula) are benchmarked against each other — and since Amani Abeid Karume runs at ~47,900 hrs, that pulls the Kondoa DC expected benchmark up to ~15,339 hrs.
- All schools in **Arusha DC** (Einoti, Losinoni, Kiranyi, Mwandet, Ilboru, etc.) share an expected benchmark of ~1,932 hrs, reflecting that most schools in that district operate in a more modest usage band.
- **Siha DC** schools (Nuru, Oshara, Namwai, Sanya Juu) share a benchmark around 9,000 hrs because that district's reference schools post strong numbers.

Expected Hours is **not** a fixed national quota and **not** based on lab capacity (number of computers) or enrollment size. It is purely a peer-performance benchmark.

3. Level — How It's Derived

Level is a simple three-tier classification based purely on the Actual hours figure:

Level	Threshold	Meaning
 Good	Actual ≥ 2,000 hrs	Healthy, regular lab utilization
 Moderate	500 ≤ Actual < 2,000 hrs	Inconsistent or partial utilization
 Poor	Actual < 500 hrs	Lab is effectively dormant or barely used

This is intentionally a simple absolute threshold rather than a relative one — it gives field teams an immediate sense of "is this lab actually being used at all" independent of what its specific peer benchmark says. A school can be **Good** in absolute terms (2,500 hrs) but still be lagging significantly behind its Expected benchmark (8,000 hrs) — Level and Gap measure two different things.

4. Gap (hrs) — How It's Derived

This is the simplest calculation in the system, but it's the foundation everything else builds on:

$$\text{Gap (hrs)} = \text{Expected (hrs)} - \text{Actual (hrs)}$$

A **positive Gap** means the school is underperforming its peer benchmark — this is what triggers inclusion in the "lagged schools" list that the Priority Rankings tab focuses on.

A **negative Gap** (shown in green in the Full Network tab) means the school is *outperforming* its district peer benchmark — these are schools doing better than expected, and they show up as Good performers or even potential "model schools" worth studying for what's working.

Example — Usongwe Secondary School:

$$\begin{array}{lcl} \text{Expected (Mbeya DC benchmark)} & = & 8,000 \text{ hrs} \\ \text{Actual} & = & 319 \text{ hrs} \\ \text{Gap} & = & 8,000 - 319 = 7,681 \text{ hrs} \end{array}$$

This 7,681-hour gap represents the raw volume of lost lab time — hours that, based on what similar schools in the same district are achieving, Usongwe's students *should* have had access to but didn't.

5. Gap ÷ Actual (Severity Multiplier) — How It's Derived and Why It Exists

This is a **severity multiplier**, and it exists specifically to correct a blind spot that raw Gap hours alone creates.

$$\text{Gap} \div \text{Actual} = \text{Gap (hrs)} \div \text{Actual (hrs)}$$

Why This Second Metric Is Necessary

Raw Gap hours favours **large absolute numbers**, which means schools in high-benchmark districts (like Kondo DC or Mbeya DC, where the Expected figure is large) will always show the biggest gaps — even if their actual usage isn't catastrophically low in proportional terms.

But a school using almost nothing — say, 2 or 10 hours in an entire year — is a fundamentally different kind of problem than a school using a moderate but insufficient amount. The multiplier captures that distinction.

Example — Ilboru Secondary School (Deactivated):

```
Expected = 1,932.5 hrs
Actual    = 3.17 hrs
Gap       = 1,929.3 hrs
Gap ÷ Actual = 1,929.3 ÷ 3.17 = 608.6×
```

That **608× multiplier** tells you something the raw 1,929-hour gap doesn't: this lab is essentially **not being used at all**. Even though its absolute gap (1,929 hrs) is smaller than Usongwe's (7,681 hrs), proportionally it is far more dormant.

Handling Near-Zero Actual Values

When Actual is extremely close to zero (below 0.5 hours, or literally zero), dividing produces an unstable or undefined number. The system caps this at a flagged value of **999×** to indicate "essentially infinite severity — this lab has had no meaningful usage" without producing a mathematically meaningless result like dividing by zero.

6. How These Four Metrics Combine — The Priority Score

None of these metrics are used in isolation for ranking. The app combines **Gap (hrs)** and **Gap ÷ Actual** into a single weighted **Priority Score**, which is what actually determines the P1–P4 tier:

Priority Score = (Gap_normalized × 60%) + (Multiplier_normalized × 40%)

Where both components are normalized to a 0–100 scale relative to the maximum value across all lagged schools, so neither metric's raw units dominate the formula.

Weight	Component	What It Rewards
60%	Absolute Gap Hours	Schools with the largest raw volume of lost lab time — high stakes in absolute terms
40%	Gap ÷ Actual Multiplier	Schools that are proportionally the most dormant — even if their absolute numbers look smaller

Why 60/40 and Not 50/50

The heavier weighting toward absolute Gap hours reflects Reneal's operational reality: a school losing 13,000 hours of potential lab access represents a much larger recoverable opportunity — in terms of total student-hours of digital literacy — than a small rural school whose entire benchmark is only 1,500 hours. The 60% weighting ensures the system doesn't over-prioritize small schools with extreme multipliers at the expense of large schools with genuinely massive absolute losses. The 40% weighting on the multiplier still ensures near-totally-dormant labs don't get buried under schools with bigger numbers but comparatively less severe proportional neglect.

Resulting Tiers

Priority Score ≥ 60	→ P1 Urgent	(0–6 month action plan)
Priority Score ≥ 35	→ P2 High	(6–12 month structured plan)
Priority Score ≥ 12	→ P3 Medium	(1–2 year monitored plan)
Priority Score < 12	→ P4 Monitor	(annual check-in only)

7. Putting It All Together — A Worked Example

Let's trace **Olokii Secondary School** through the entire pipeline:

Step	Value	Source
Most recent usage year available	2025	Raw usage logs
Actual	146.75 hrs	Direct from 2025 log
District	Arusha DC	School metadata
Expected	1,932.5 hrs	Arusha DC peer benchmark (average of district reference schools)
Level	Poor	$146.75 < 500$ threshold
Gap	$1,932.5 - 146.75 = \mathbf{1,785.8}$ hrs	Expected – Actual
Gap ÷ Actual	$1,785.8 \div 146.75 = \mathbf{12.17\times}$	Gap ÷ Actual
Priority Score	(Gap normalized $\times 0.6$) + (Multiplier normalized $\times 0.4$)	Weighted combination
Tier	P3 Medium	Based on the resulting score band

Olokii's relatively modest absolute gap (compared to schools like Usongwe at 7,681 hrs) combined with its moderate multiplier (12.17 $\times$ — concerning but not extreme like Ilboru's 608 $\times$) places it in P3 Medium rather than P1 or P2 — meaning it warrants monitoring and a structured plan, but it isn't flagged for the most urgent immediate field deployment.

Why This Design Matters for Reneal's Cross-Schools

This four-metric structure gives Reneal a defensible, transparent answer to three different stakeholders simultaneously: (Gap hours), (Level + regional grouping), and the field operations team gets a single actionable number (Priority Score) telling them where to go first — all derived from real usage logs rather than subjective field judgment, with human assessment left to the team members on-the-ground assessment rather than baked into the formula.

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